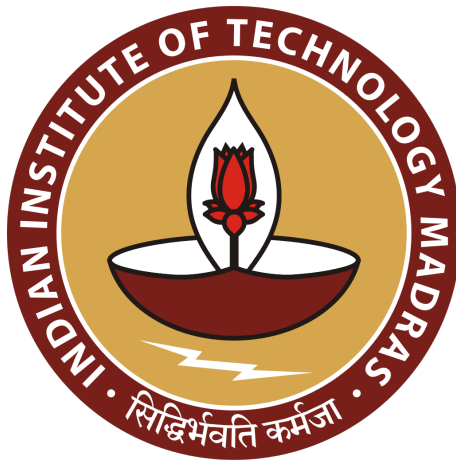


Assignment -2

Regression and Classification



BT3041 - Analysis and Interpretation of Biological Data

Even Semester 2026

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Question 1

Part (a)

The data was standardized (i.e. Z-score Normalization was performed) and was then split into training, validation, and test datasets in a 60:20:20 ratio, using the `np.split` function as shown below:

```
train, val, test = np.split(df.sample(frac=1, random_state = 67),
                               [int(.6*len(df)), int(.8*len(df))])
```

The ratio of split was confirmed by checking the size of the split datasets, as shown in the output below:

```
The size of train data is: 994
The size of validation data is: 332
The size of test data is: 332
```

Following this the features and (labels/target) variables are separated accordingly.

NOTE: Z-score Normalization was preferred over Min-Max Normalization, since no check was performed for outliers and I wanted the analysis to be robust against outliers.

Part (b)

A power-set of all possible subsets of the 14 features from the training dataset was constructed, and the RSS vs number of features plot is shown below:

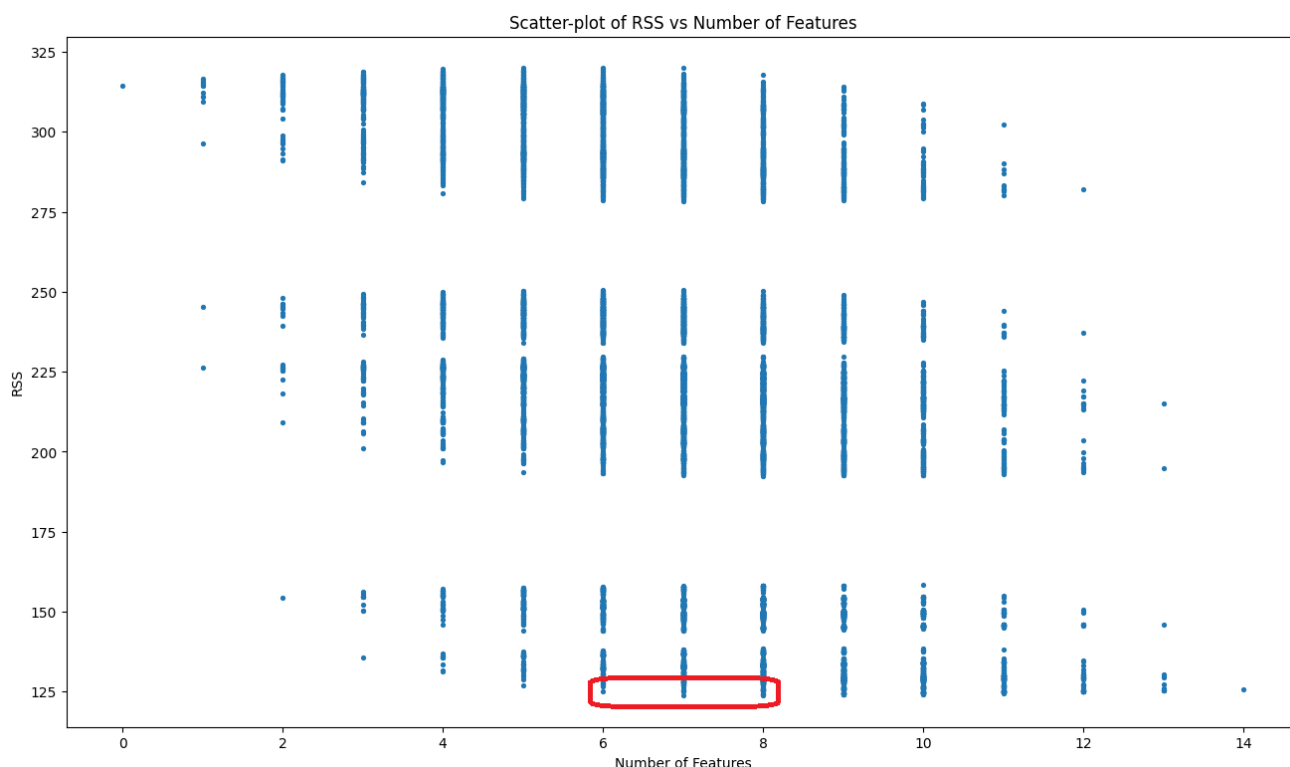


Figure 1.b.1: Scatter-plot of RSS vs Number of Features

From the above plot it becomes evident that the model that minimizes the RSS and AIC, would have anywhere between 6 to 8 features.

The table below shows the top 3 models (sorting on the basis of AIC), with their corresponding RSS values.

S.No	Number of Features	Features	RSS	AIC
1	7	N, P, K, Soil_Moisture, Rainfall, Wind_Speed, Fertilizer_Used	123.824044	1937.725361
2	8	N, P, K, Soil_Moisture, Rainfall, Wind_Speed, Altitude, Fertilizer_Used	124.234687	1937.954318
3	8	N, P, K, Soil_Moisture, Organic_Carbon, Rainfall, Wind_Speed, Fertilizer_Used	124.088363	1938.395715

The above table clearly shows that the eighth parameter only marginally improves performance and thus, the first model is the simplest model that captures our dataset accurately (However, to ensure that this model is not biased by specific patterns in the training data, we perform 5-fold cross-validation, in Part (c)).

Part (c)

5-fold cross-validation was performed on the top 3 models in Part (b) (using the training dataset) to assess its stability. The results are shown below:

```

Features: ['N', 'P', 'K', 'Soil_Moisture', 'Rainfall', 'Wind_Speed',
'Fertilizer_Used']
Mean MSE: 0.41735935856849793
Variance MSE: 0.00016489142567142074
Standard Deviation of MSE: 0.012841005633182346
=====
Features: ['N', 'P', 'K', 'Soil_Moisture', 'Rainfall', 'Wind_Speed',
'Altitude', 'Fertilizer_Used']
Mean MSE: 0.41735935856849793
Variance MSE: 0.00016489142567142074
Standard Deviation of MSE: 0.012841005633182346
=====
Features: ['N', 'P', 'K', 'Soil_Moisture', 'Organic_Carbon', 'Rainfall',
'Wind_Speed', 'Fertilizer_Used']
Mean MSE: 0.41735935856849793
Variance MSE: 0.00016489142567142074
Standard Deviation of MSE: 0.012841005633182346
=====

```

All 3 models show a low standard deviation, and have low MSE values, this indicates that these models have not been biased by specific patterns in the training data.

Additionally this now validates that fact that the first model is clearly the best, as it is the simplest model that accurately captures the key trends in our training dataset.

Part (d)

The metrics RMSE, MAE, and R2 of the top 3 models' was computed using the validation dataset and are shown below:

```

For model number 1:
The Coefficient of Determination (R2) is 0.425
The Mean Absolute Error (MAE) is 0.513

```

```
The Mean Squared Error (MSE) is 0.396
The Root Mean Squared Error (RMSE) is 0.629
=====
```

```
For model number 2:
The Coefficient of Determination (R2) is 0.425
The Mean Absolute Error (MAE) is 0.513
The Mean Squared Error (MSE) is 0.396
The Root Mean Squared Error (RMSE) is 0.629
=====
```

```
For model number 3:
The Coefficient of Determination (R2) is 0.421
The Mean Absolute Error (MAE) is 0.515
The Mean Squared Error (MSE) is 0.399
The Root Mean Squared Error (RMSE) is 0.632
=====
```

The above output clearly indicates that, although the top three models do not exhibit significant bias, they perform poorly in predicting the system output. The low R^2 value across all the models is indicative of this fact. The low RMSE and MSE seen here are due to the fact that the data has been standardized, even on this metric all 3 models perform more or less the same, further indicating that the inclusion of the eighth parameter does not meaningfully improve the model.

Part (e)

L1 and L2 regularisation-based regression models were constructed using the training data and $\lambda = 0.25$ (for both regularisation models).

The coefficients and intercepts obtained in each case are shown below:

L1 regularisation:

```
The coefficient beta_0 is: -0.019511765678682243
The coefficient beta_1 is: 0.0
The coefficient beta_2 is: 0.0
The coefficient beta_3 is: 0.0
The coefficient beta_4 is: -0.0
The coefficient beta_5 is: 0.024995682190793907
The coefficient beta_6 is: -0.0
The coefficient beta_7 is: -0.0
The coefficient beta_8 is: 0.0
The coefficient beta_9 is: 0.2027567313650134
The coefficient beta_10 is: 0.0
The coefficient beta_11 is: -0.0
The coefficient beta_12 is: 0.0
The coefficient beta_13 is: 0.2833181294060915
The coefficient beta_14 is: 0.0
```

- We clearly see that Lasso Regression does feature selection, a bunch of weights have been set to zero, indicating these features are not correlated strongly with the target variable.
- We see that the 9th and 13th parameters have a large positive coefficient. These are "Fertilizer_Used" and "Rainfall" respectively.
- From the above we can infer that "Fertilizer_Used" and "Rainfall" are most strongly positively correlated with increased Crop_Yield.

L2 regularisation:

```
The coefficient beta_0 is: -0.008686043971882645
The coefficient beta_1 is: 0.13836193835395716
The coefficient beta_2 is: 0.10312662370008151
The coefficient beta_3 is: 0.13133337449237747
The coefficient beta_4 is: -0.011441070262838448
The coefficient beta_5 is: 0.2708870941468055
The coefficient beta_6 is: -0.02474185082896703
The coefficient beta_7 is: -0.006237189400727315
The coefficient beta_8 is: 0.018793849722226013
The coefficient beta_9 is: 0.41596447834432737
The coefficient beta_10 is: 0.014536889909192892
The coefficient beta_11 is: -0.17072436138347086
The coefficient beta_12 is: 0.026142791029285763
The coefficient beta_13 is: 0.5119441458426832
The coefficient beta_14 is: 0.013429731028574891
```

- We clearly see that Ridge Regression does not perform feature selection. However, it does shrink coefficients towards zero.
- We see that the 5th, 9th and 13th parameters have a large positive coefficient. These are "Soil_Moisture", "Fertilizer_Used" and "Rainfall" respectively.
- Features with near-zero weights may arise either due to Ridge Regression mitigating multicollinearity, or because the feature itself is not strongly correlated with Crop_Yield.

Part (f)

5-fold cross-validation was performed on the training dataset to fine-tune the λ value of Lasso and Ridge regression respectively. The output is shown below:

Lasso Regression:

```
Best model searched:
lambda = 0.007245131724078188
intercept = -0.009023516923541738,
betas = [ 1.30202396e-01  9.52962946e-02  1.24876217e-01 -4.06757534e-03
          2.64120514e-01 -1.84221908e-02 -2.89906883e-04  1.14083673e-02
          4.10688030e-01  7.87832387e-03 -1.63979026e-01  1.94855113e-02
          5.05606979e-01  6.37742572e-03]
```

Ridge Regression:

```
Best model searched:
lambda = 10.0
intercept = -0.00884289774708688,
betas = [ 0.13738175  0.10192423  0.12989507 -0.01117551  0.2681881 -0.02497404
          -0.00661661  0.01844922  0.4124647  0.01453279 -0.16929641  0.0258946
          0.5072099  0.01339169]
```

The optimal value of λ was obtained for both regularisation techniques and the corresponding models were created and validated in the next part of the question.

Part (g)

The metrics RMSE, MAE, and R2 for both the regularisation models was computed and are shown below.

Lasso Regression:

```
The coefficient beta_0 is: -0.009023516923541738
The coefficient beta_1 is: 0.1302023963450758
The coefficient beta_2 is: 0.09529629462900231
The coefficient beta_3 is: 0.1248762165261716
The coefficient beta_4 is: -0.004067575343630665
The coefficient beta_5 is: 0.26412051421415256
The coefficient beta_6 is: -0.018422190841430174
The coefficient beta_7 is: -0.0002899068828994571
The coefficient beta_8 is: 0.011408367285936576
The coefficient beta_9 is: 0.41068802993458575
The coefficient beta_10 is: 0.007878323865956206
The coefficient beta_11 is: -0.1639790260014284
The coefficient beta_12 is: 0.019485511303699063
The coefficient beta_13 is: 0.5056069793258826
The coefficient beta_14 is: 0.006377425716297297
```

```
=====
The Coefficient of Determination (R2) is 0.637
The Mean Absolute Error (MAE) is 0.462
The Mean Squared Error (MSE) is 0.336
The Root Mean Squared Error (RMSE) is 0.58
=====
```

Ridge Regression:

```
The coefficient beta_0 is: -0.00884289774708688
The coefficient beta_1 is: 0.13738175421295828
The coefficient beta_2 is: 0.10192422560524621
The coefficient beta_3 is: 0.12989507140349063
The coefficient beta_4 is: -0.011175510071133342
The coefficient beta_5 is: 0.26818810294800166
The coefficient beta_6 is: -0.02497404323905867
The coefficient beta_7 is: -0.00661660997751029
The coefficient beta_8 is: 0.018449215442236665
The coefficient beta_9 is: 0.4124646962141588
The coefficient beta_10 is: 0.01453279150402752
The coefficient beta_11 is: -0.1692964097568896
The coefficient beta_12 is: 0.02589460143138087
The coefficient beta_13 is: 0.507209901271628
The coefficient beta_14 is: 0.013391690141414978
```

```
=====
The Coefficient of Determination (R2) is 0.637
The Mean Absolute Error (MAE) is 0.462
The Mean Squared Error (MSE) is 0.336
The Root Mean Squared Error (RMSE) is 0.58
=====
```

In terms of performance on the validation dataset, both the regularisation models appear to be equally good, and far better than the previous linear regression models (atleast in terms of R^2 and RMSE).

However, we will need to compare the model on test data (executed in Part (h)) to make any decisive statements.

Part (h)

The metrics RMSE, MAE, and R^2 of the top 3 models' was computed using the test dataset for all the best models and are shown below:

Top 3 Linear Regression models:

```
For model number 1:
The Coefficient of Determination (R2) is 0.602
The Mean Absolute Error (MAE) is 0.488
The Mean Squared Error (MSE) is 0.373
The Root Mean Squared Error (RMSE) is 0.611
```

```
=====
For model number 2:
The Coefficient of Determination (R2) is 0.601
The Mean Absolute Error (MAE) is 0.489
The Mean Squared Error (MSE) is 0.374
The Root Mean Squared Error (RMSE) is 0.612
```

```
=====
For model number 3:
The Coefficient of Determination (R2) is 0.601
The Mean Absolute Error (MAE) is 0.488
The Mean Squared Error (MSE) is 0.374
The Root Mean Squared Error (RMSE) is 0.612
=====
```

Lasso Regression:

```
The Coefficient of Determination (R2) is 0.6
The Mean Absolute Error (MAE) is 0.489
The Mean Squared Error (MSE) is 0.375
The Root Mean Squared Error (RMSE) is 0.612
=====
```

Ridge Regression:

```
The Coefficient of Determination (R2) is 0.596
The Mean Absolute Error (MAE) is 0.491
The Mean Squared Error (MSE) is 0.379
The Root Mean Squared Error (RMSE) is 0.616
=====
```

Part (i)

- The three linear regression models exhibit nearly identical performance, with very minimal differences in R^2 , MAE, MSE, and RMSE. This indicates that the seven parameters that maximize predictive capability significantly are [N, P, K, Soil_Moisture, Rainfall, Wind_Speed, Fertilizer_Used].

- Lasso Regression achieves performance comparable to the best linear models. However, it is curious to note that in Part(g), there has been no effective feature selection by Lasso Regression, as it has opted to keep all the parameters with minimal weights at the least.
- Out of all the models Ridge Regression is the worst performing (though not by a large margin). It shows slightly lower R^2 and higher error metrics compared to the other models.
- The similar performance observed across all models suggests the presence of multi-collinearity within the feature variables. As a result, increasing model complexity or applying regularization does not lead to substantial improvements in predictive accuracy.

To maximize the crop yield the farmer would need to ensure that:

- The soil is fertilized sufficiently (Most important)
- Cultivation of the crop takes place during the Monsoon Season
- Nitrogen and Potassium content, and Soil Moisture must be maintained optimally.

Question 2

Part (a)

The categorical data was converted into continuous data using the maps given in the question. The feature data was standardized (i.e. Z-score Normalization was performed), while the target labels were left untouched here. Following this the data was split into training, validation, and test datasets in a 60:20:20 ratio, using the np.split function as shown below:

```
train, val, test = np.split(df.sample(frac=1, random_state = 67),
                             [int(.6*len(df)), int(.8*len(df))])
```

The ratio of split was confirmed by checking the size of the split datasets, as shown below:

```
The size of train data is: 994
The size of validation data is: 332
The size of test data is: 332
```

Following this the features and (labels/target) variables are separated accordingly.

NOTE: Z-score Normalization was preferred over Min-Max Normalization, since no check was performed for outliers and I wanted the analysis to be robust against outliers.

Part (b)

A Decision Tree was created using the training dataset as shown below:

```
feature_cols = ['N', 'P', 'K', 'Soil_pH', 'Soil_Moisture', 'Organic_Carbon',
                'Temperature', 'Humidity', 'Rainfall', 'Sunlight_Hours', 'Wind_Speed',
                'Altitude', 'Fertilizer_Used', 'Pesticide_Used', 'Soil_Type',
                'Irrigation_Type']

clf = DecisionTreeClassifier(criterion='entropy', random_state=42)

clf_1 = clf.fit(X_train, y_train)
```

It was evaluated using validation data and its confusion matrix is shown below:

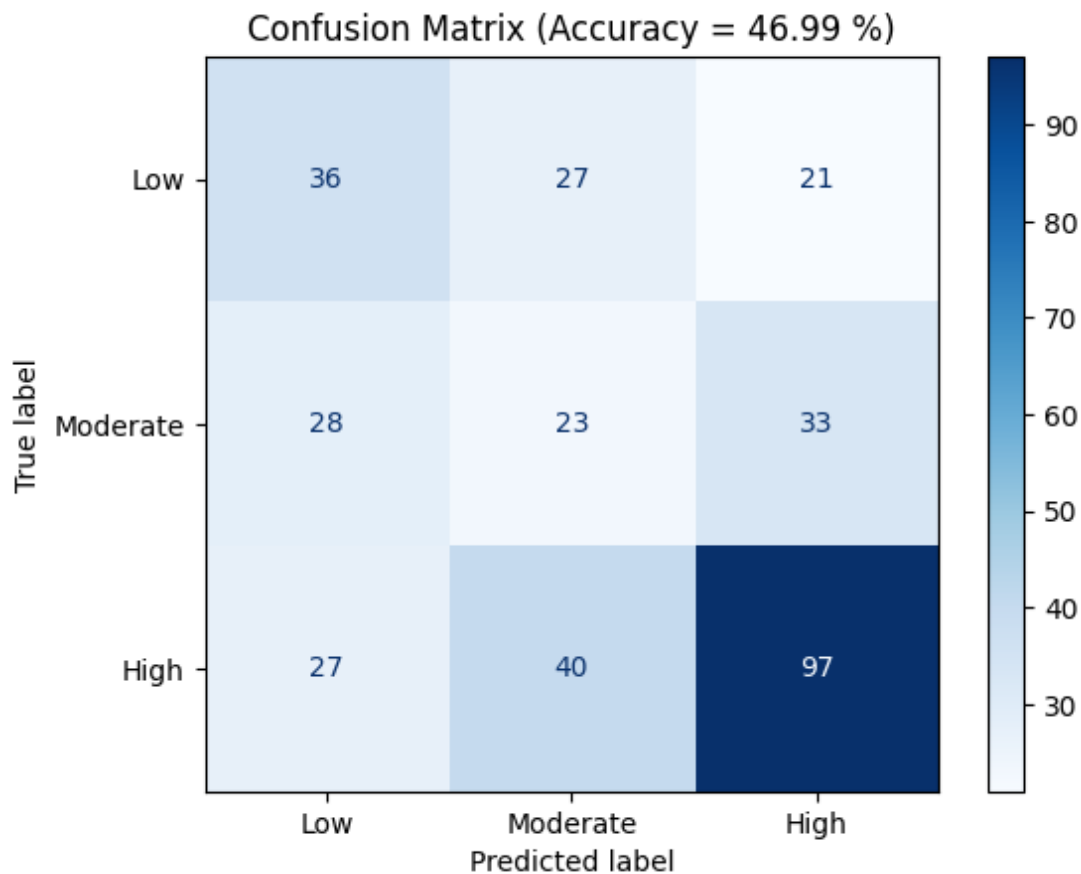


Figure 2.b.1: Confusion Matrix for the above Decision Tree

As it can be clearly seen, this Decision Tree has very poor predicting capability.

Part (c)

5-fold cross-validation was performed on the training dataset for the decision tree classifier to determine the optimal tree depth that maximizes the model's classification capability.

Since the data in this case is skewed toward one of the classes (more than 50% of the data belongs to "High" yield class), both "accuracy" and "balanced accuracy" were used as scoring metrics to determine the optimal Decision Tree. .

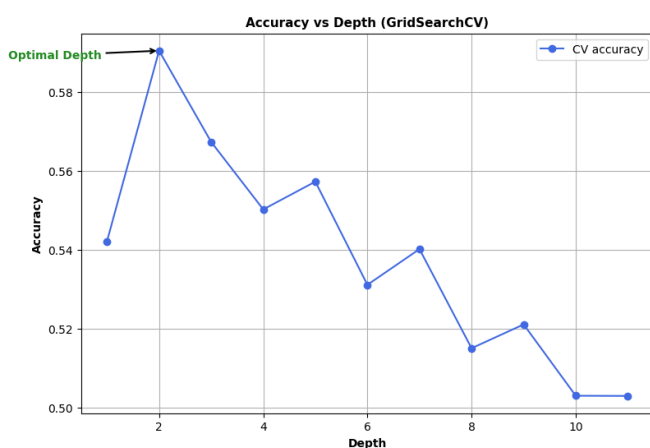


Figure 2.c.1: Accuracy vs Depth curve

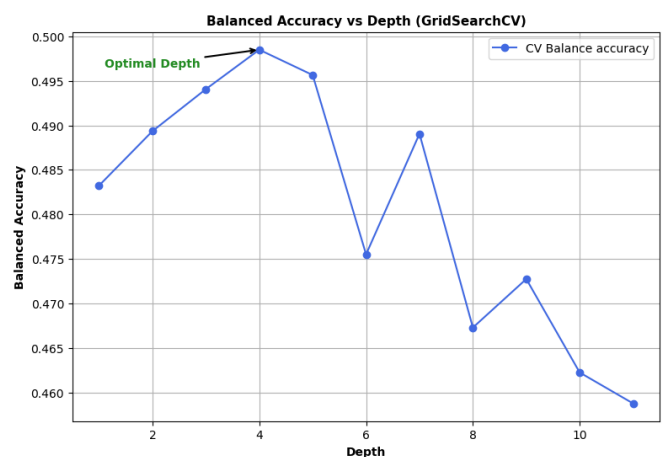


Figure 2.c.2: Balanced Accuracy vs Depth curve

Accuracy (scoring metric)

From Figure 2.c.1 we deduce that a depth of 2 is most optimal for the Decision tree (using accuracy as the scoring metric).

Balanced Accuracy (scoring metric)

From Figure 2.c.2 we deduce that a depth of 4 is most optimal for the Decision tree (using balanced accuracy as the scoring metric).

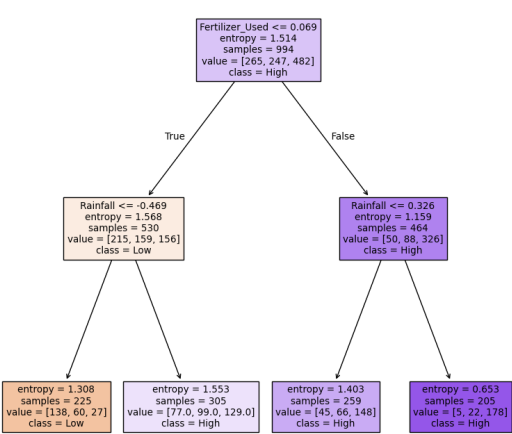


Figure 2.c.3: Visualisation of the tree constructed from accuracy based scoring

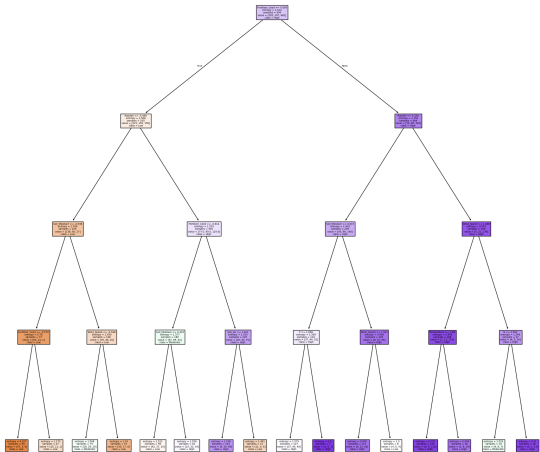


Figure 2.c.4: Visualisation of the tree constructed from accuracy based scoring

Part (d)

The Decision Tree’s metrics (confusion matrix, precision, recall, F1-score (micro and macro averaged), ROC curve, and AUC) were evaluated using the validation dataset, and the results are shown below:

Accuracy (scoring metric)

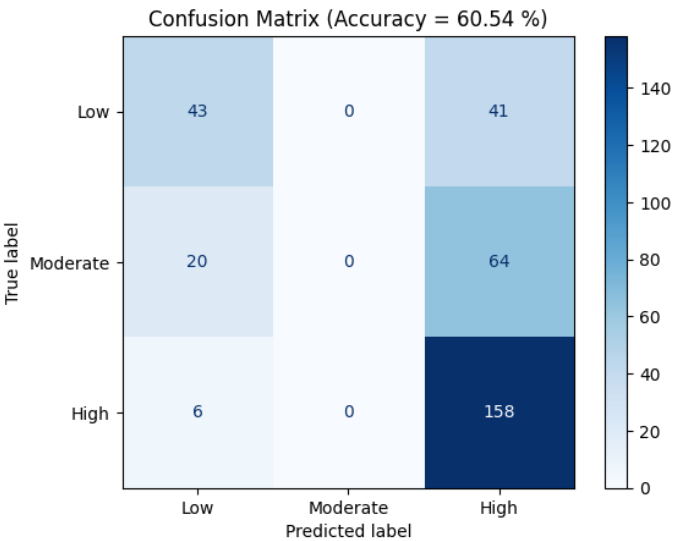


Figure 2.d.1: Confusion Matrix

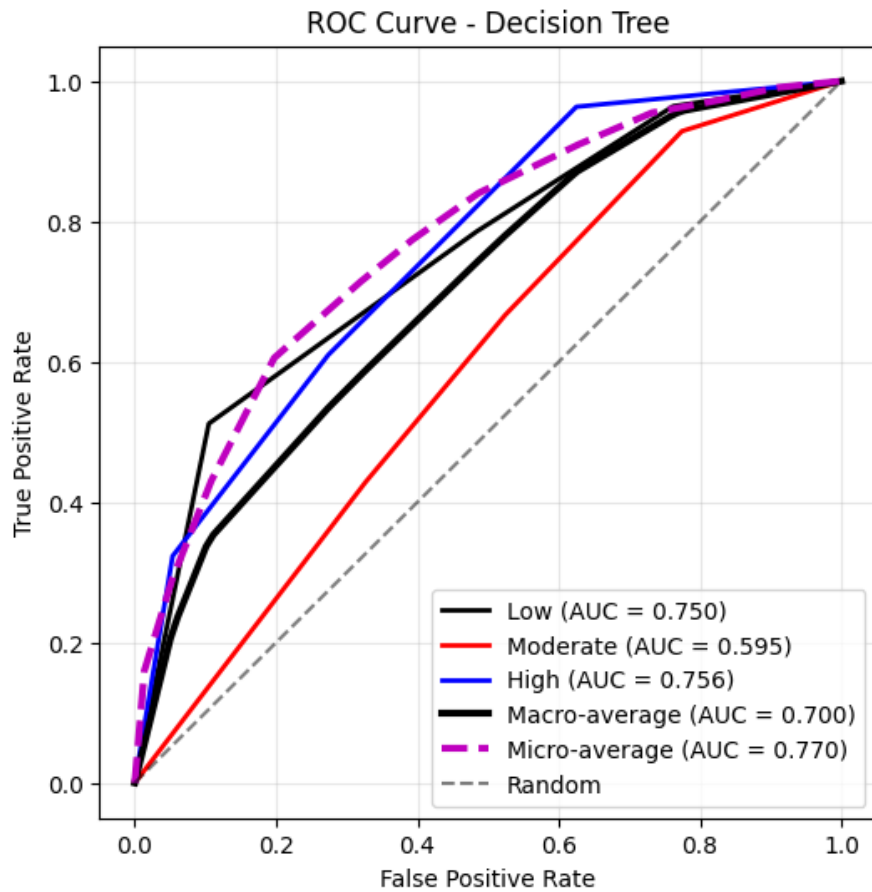


Figure 2.d.2: ROC curve

Type of Average	Precision	Recall	F1-score
Micro-averaged	0.6054	0.6054	0.6054
Macro-averaged	0.41	0.49	0.51

Balanced Accuracy (scoring metric)

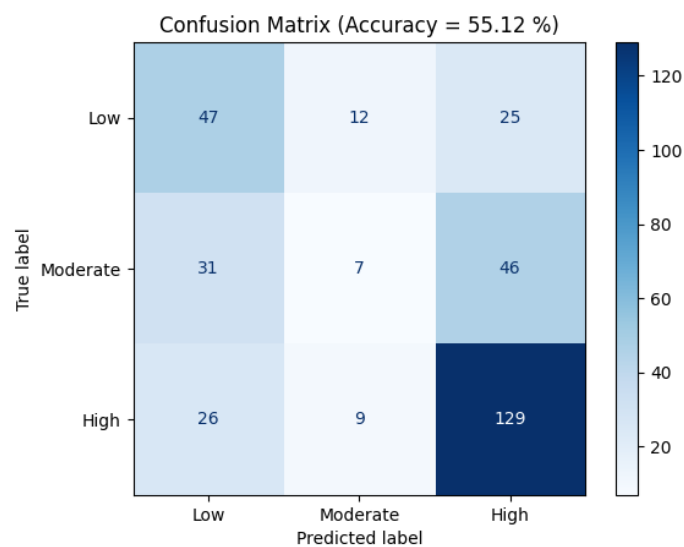


Figure 2.d.3: Confusion Matrix

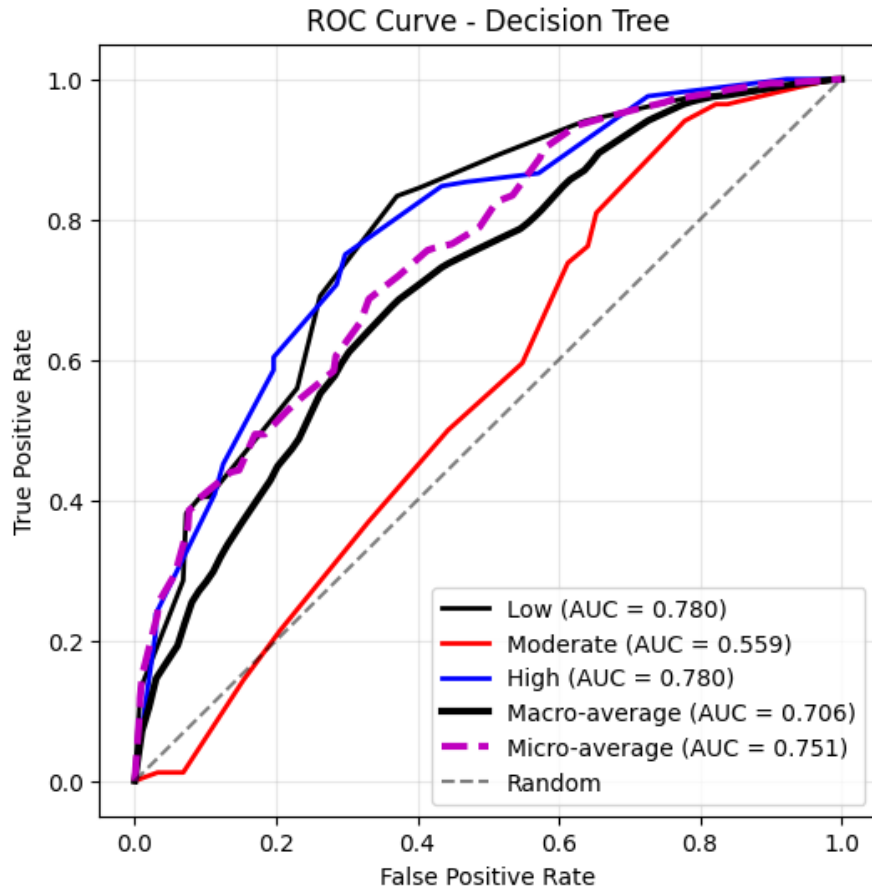


Figure 2.d.4: ROC curve

Type of Average	Precision	Recall	F1-score
Micro-averaged	0.5512	0.5512	0.5512
Macro-averaged	0.45	0.48	0.44

- In the first case (depth = 2), the model primarily classifies extreme classes, showing a strong bias toward the "High" class due to data imbalance, with no predictions for the "Moderate" class (Figures 2.c.3 and 2.d.1).
- In the second case (depth = 4), the model begins to capture minority classes, but at the cost of reduced accuracy in classifying the "High" class.
- In both cases, the model performs poorly on the "Moderate" class, as indicated by an AUC close to 0.5, suggesting performance only marginally better than random guessing.

Part (e)

We build kNN models and determine the optimal value of k by fitting the model on the training data for different values of k , and analyzing a plot of error vs k value.

From Figure 2.e.1, we see that for k belongs to (20,65) the error rate only reduces marginally.

Although the graph identifies an optimal value at $k = 61$, we cannot conclude that this choice does not lead to overfitting. Therefore, a cross-validation analysis is necessary to assess model stability before selecting a truly optimal value of k .

Given below is the plot of Error score vs k value:

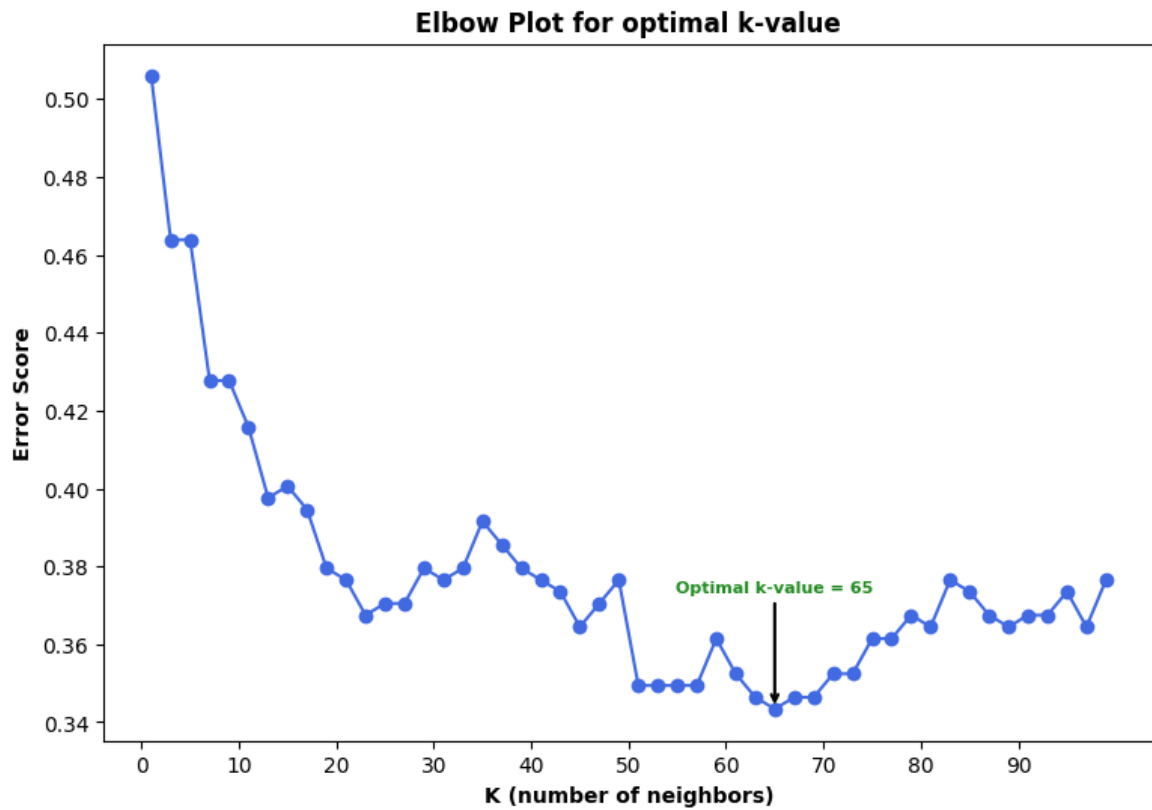


Figure 2.e.1: Elbow plot to determine optimal k

Part (f)

A 5-fold cross-validation was performed using the training dataset, where k belongs to (20,70).

The plots for average error score vs k and standard deviation in scores vs k is shown below:

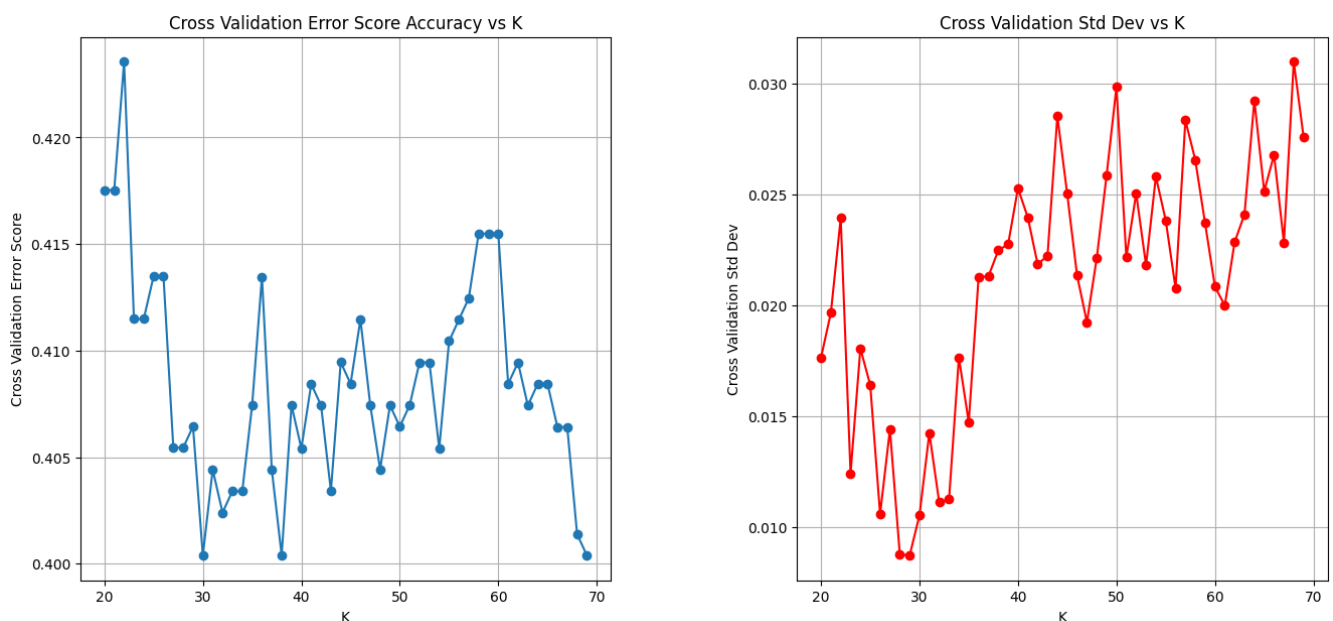


Figure 2.f.1: Determining optimal k using 5-fold cross-validation

Now, it becomes apparent that the optimal value of $k \approx 30$, as this value minimizes both the mean error score and its variance, indicating that the model effectively captures the underlying patterns in the training data without over/under-fitting.

Part (g)

The kNN's metrics (confusion matrix, precision, recall, F1-score (micro and macro averaged), ROC curve, and AUC) were evaluated using the validation dataset, and the results are shown below:

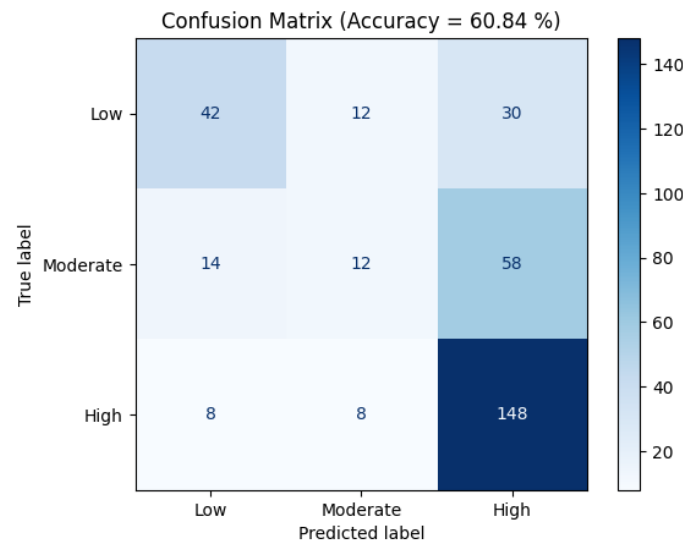


Figure 2.g.1: Confusion Matrix

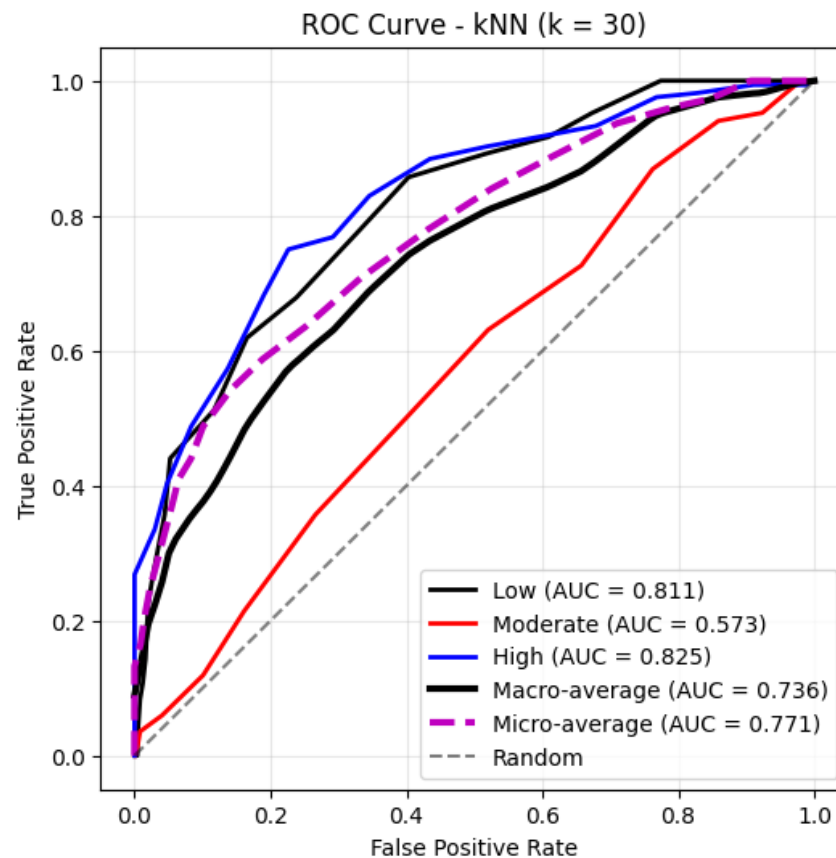


Figure 2.g.2: ROC curve

Type of Average	Precision	Recall	F1-score
Micro-averaged	0.6084	0.6084	0.6084
Macro-averaged	0.55	0.52	0.50

- The class imbalance seen in the Decision-Trees persists, with only marginal improvement, as reflected by the slightly higher AUC for the "Moderate" class.
- This model however, has a higher accuracy score than both the Decision-Tress and doesn't outright ignore minority classes.

Part (h)

Decision Tree classifier

The accuracy and confusion matrix was determined for the best models in each category using the test dataset and the results are shown below:

Accuracy (scoring metric)

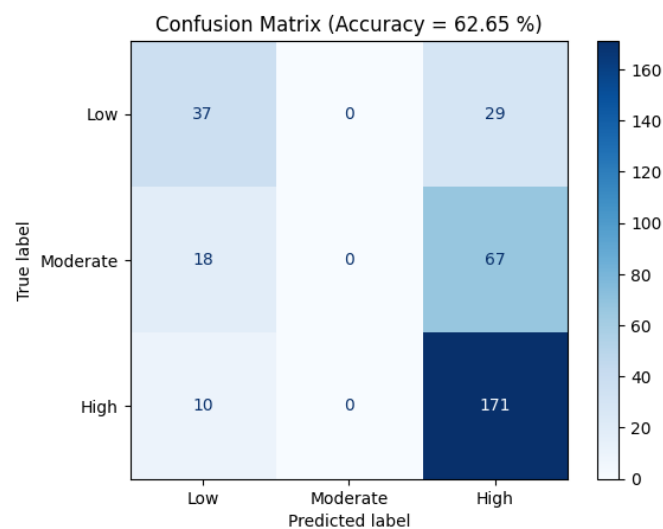


Figure 2.h.1: Confusion Matrix (Tree depth = 2)

Balanced Accuracy (scoring metric)

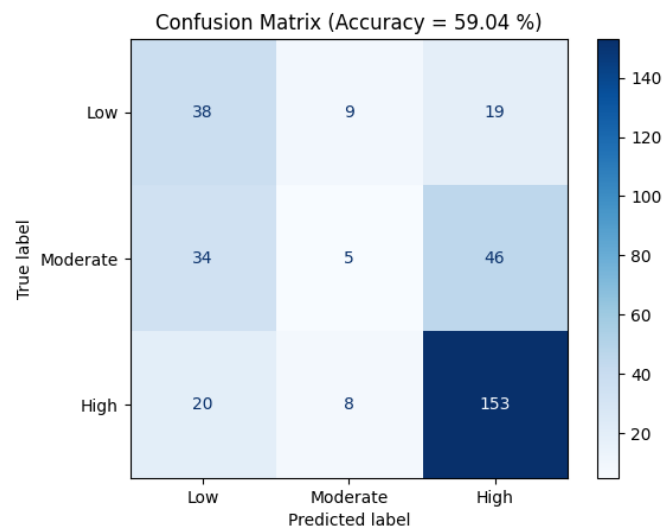


Figure 2.h.2: Confusion Matrix (Tree depth = 4)

k-Nearest Neighbours classifier

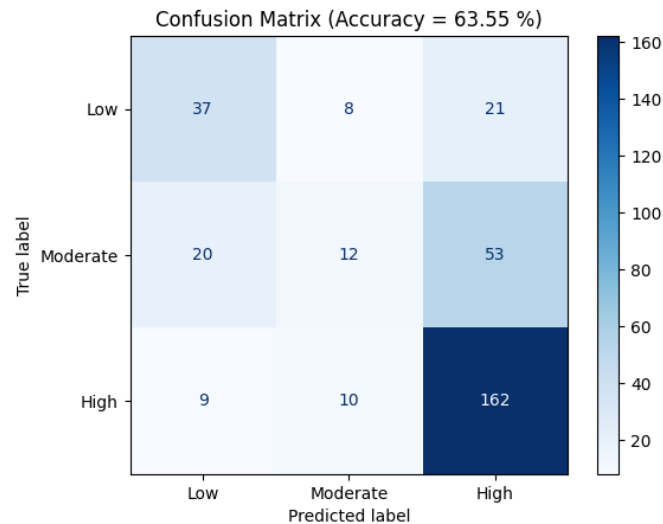


Figure 2.h.3: Confusion Matrix (kNN, k = 30)

Conclusions

In terms of pure accuracy values:

kNN (k = 30) > Decision-Tree (depth = 2) > Decision-Tree (depth = 4)

63.5 % > 62.65 % > 59.04%

There are some caveats to be considered before drawing a final conclusion:

- The Decision-Tree (depth = 2) cannot be considered a good classifier, since it actively ignore the "Moderate" class from it's predictions. As such it cannot be considered the best model, and I would go so far as to say it cannot be considered to be better than the Decision-Tree (depth = 4)
- The Decision-Tree (depth = 4) can also not be considered a great classifier, though arguably better than the Decision-Tree (depth = 2), it makes far to many errors in predicting the "High" class accurately.
- The kNN model (k = 30) occupies a Goldilocks zone, achieving a balanced trade-off where it retains strong predictive performance for the "High" class while also providing reasonable classification of the minority classes.

Therefore, I conclude that the kNN classifier is the most suitable model for this dataset, not only based on its accuracy but also considering the above mentioned factors.

Link to the google drive containing the python code used in this assignment:

https://drive.google.com/drive/u/0/folders/1SV1UNDINONa0DhPSUT8HEwh7_7UtquP0